

9

Model Uncertainty and Robustness

In this chapter we briefly describe various types of uncertainties which can arise in physical systems, and we single out “unstructured uncertainties” as generic errors which are associated with all design models. We obtain robust stability tests for systems under various model uncertainty assumptions through the use of the small gain theorem. And we also obtain some sufficient conditions for robust performance under unstructured uncertainties. The difficulty associated with MIMO robust performance design and the role of plant condition numbers for systems with skewed performance and uncertainty specifications are revealed. We show by examples that the classical gain margin and phase margin are insufficient indicators for the system robustness. A simple example is also used to indicate the fundamental difference between the robustness of an SISO system and that of an MIMO system. In particular, we show that applying the SISO analysis/design method to an MIMO system may lead to erroneous results.

9.1 Model Uncertainty

Most control designs are based on the use of a design model. The relationship between models and the reality they represent is subtle and complex. A mathematical model provides a map from inputs to responses. The quality of a model depends on how closely its responses match those of the true plant. Since no single fixed model can respond exactly like the true plant, we need, at the very least, a set of maps. However, the

modeling problem is much deeper – the universe of mathematical models from which a model set is chosen is distinct from the universe of physical systems. Therefore, a model set which includes the true physical plant can never be constructed. It is necessary for the engineer to make a leap of faith regarding the applicability of a particular design based on a mathematical model. To be practical, a design technique must help make this leap small by accounting for the inevitable inadequacy of models. A good model should be simple enough to facilitate design, yet complex enough to give the engineer confidence that designs based on the model will work on the true plant.

The term *uncertainty* refers to the differences or errors between models and reality, and whatever mechanism is used to express these errors will be called a *representation of uncertainty*. Representations of uncertainty vary primarily in terms of the amount of structure they contain. This reflects both our knowledge of the physical mechanisms which cause differences between the model and the plant and our ability to represent these mechanisms in a way that facilitates convenient manipulation. For example, consider the problem of bounding the magnitude of the effect of some uncertainty on the output of a nominally fixed linear system. A useful measure of uncertainty in this context is to provide a bound on the spectrum of the output's deviation from its nominal response. In the simplest case, this spectrum is assumed to be independent of the input. This is equivalent to assuming that the uncertainty is generated by an additive noise signal with a bounded spectrum; the uncertainty is represented as additive noise. Of course, no physical system is linear with additive noise, but some aspects of physical behavior are approximated quite well using this model. This type of uncertainty received a great deal of attention in the literature during the 1960's and 1970's, and the attention is probably due more to the elegant theoretical solutions that are yielded (e.g., white noise propagation in linear systems, Wiener and Kalman filtering, LQG) than to the great practical significance offered.

Generally, the deviation's spectrum of the true output from the nominal will depend significantly on the input. For example, an additive noise model is entirely inappropriate for capturing uncertainty arising from variations in the material properties of physical plants. The actual construction of model sets for more general uncertainty can be quite difficult. For example, a set membership statement for the parameters of an otherwise known FDLTI model is a highly-structured representation of uncertainty. It typically arises from the use of linear incremental models at various operating points, e.g., aerodynamic coefficients in flight control vary with flight environment and aircraft configurations, and equation coefficients in power plant control vary with aging, slag buildup, coal composition, etc. In each case, the amounts of variation and any known relationships between parameters can be expressed by confining the parameters to appropriately defined subsets of parameter space. However, for certain classes of signals (e.g., high frequency), the parameterized FDLTI model fails to describe the plant because the plant will always have dynamics which are not represented in the fixed order model.

In general, we are forced to use not just a single parameterized model but model sets that allow for plant dynamics which are not explicitly represented in the model structure.

A simple example of this involves using frequency-domain bounds on transfer functions to describe a model set. To use such sets to describe physical systems, the bounds must roughly grow with frequency. In particular, at sufficiently high frequencies, phase is completely unknown, i.e., $\pm 180^\circ$ uncertainties. This is a consequence of dynamic properties which inevitably occur in physical systems. This gives a less structured representation of uncertainty.

Examples of less-structured representations of uncertainty are direct set membership statements for the transfer function matrix of the model. For instance, the statement

$$P_\Delta(s) = P(s) + W_1(s)\Delta(s)W_2(s), \quad \bar{\sigma}[\Delta(j\omega)] < 1, \quad \forall \omega \geq 0, \quad (9.1)$$

where W_1 and W_2 are stable transfer matrices that characterize the spatial and frequency structure of the uncertainty, confines the matrix P_Δ to a neighborhood of the nominal model P . In particular, if $W_1 = I$ and $W_2 = w(s)I$ where $w(s)$ is a scalar function, then P_Δ describes a disk centered at P with radius $w(j\omega)$ at each frequency as shown in Figure 9.1. The statement does not imply a mechanism or structure which gives rise to Δ . The uncertainty may be caused by parameter changes, as mentioned above or by neglected dynamics or by a host of other unspecified effects. An alternative statement to (9.1) is the so-called multiplicative form:

$$P_\Delta(s) = (I + W_1(s)\Delta(s)W_2(s))P(s). \quad (9.2)$$

This statement confines P_Δ to a normalized neighborhood of the nominal model P . An advantage of (9.2) over (9.1) is that in (9.2) compensated transfer functions have the same uncertainty representation as the raw model (i.e., the weighting functions apply to PK as well as P). Some other alternative set membership statements will be discussed later.

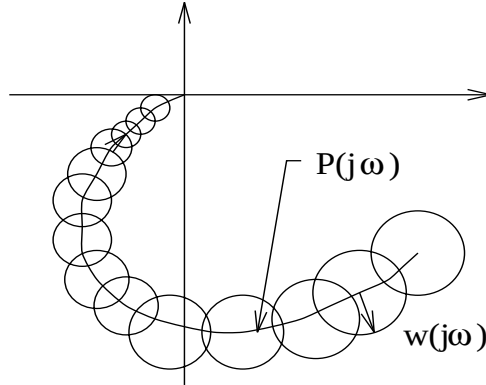


Figure 9.1: Nyquist Diagram of an Uncertain Model

The best choice of uncertainty representation for a specific FDLTI model depends, of course, on the errors the model makes. In practice, it is generally possible to represent

some of these errors in a highly-structured parameterized form. These are usually the low frequency error components. There are always remaining higher frequency errors, however, which cannot be covered this way. These are caused by such effects as infinite-dimensional electro-mechanical resonance, time delays, diffusion processes, etc. Fortunately, the less-structured representations, e.g., (9.1) or (9.2), are well suited to represent this latter class of errors. Consequently, (9.1) and (9.2) have become widely used “generic” uncertainty representations for FDLTI models. An important point is that the construction of the weighting matrices W_1 and W_2 for multivariable systems is not trivial.

Motivated from these observations, we will focus for the moment on the multiplicative description of uncertainty. We will assume that P_Δ in (9.2) remains a strictly proper FDLTI system for all Δ . More general perturbations (e.g., time varying, infinite dimensional, nonlinear) can also be covered by this set provided they are given appropriate “conic sector” interpretations via Parseval’s theorem. This connection is developed in [Safonov, 1980] and [Zames, 1966] and will not be pursued here.

When used to represent the various high frequency mechanisms mentioned above, the weighting functions in (9.2) commonly have the properties illustrated in Figure 9.2. They are small ($\ll 1$) at low frequencies and increase to unity and above at higher frequencies. The growth with frequency inevitably occurs because phase uncertainties eventually exceed ± 180 degrees and magnitude deviations eventually exceed the nominal transfer function magnitudes. Readers who are skeptical about this reality are encouraged to try a few experiments with physical devices.

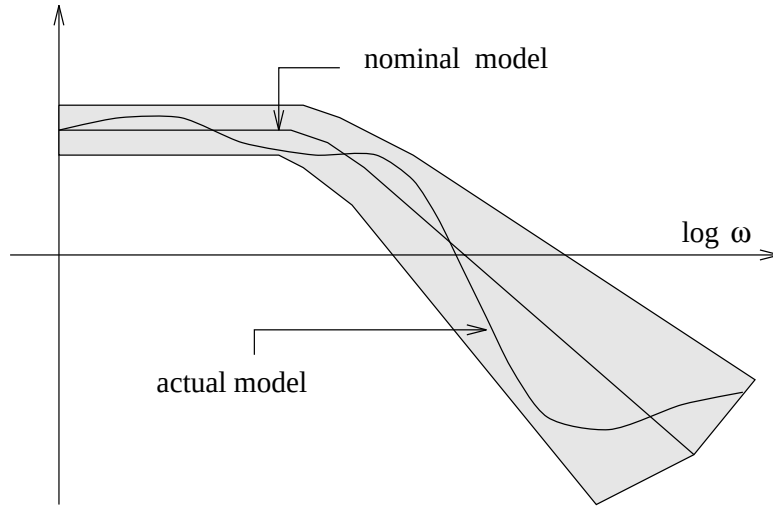


Figure 9.2: Typical Behavior of Multiplicative Uncertainty: $p_\delta(s) = [1 + w(s)\delta(s)]p(s)$

Also note that the representation of uncertainty in (9.2) can be used to include perturbation effects that are in fact certain. A nonlinear element, may be quite accurately

modeled, but because our design techniques cannot effectively deal with the nonlinearity, it is treated as a conic linearity¹. As another example, we may deliberately choose to ignore various known dynamic characteristics in order to achieve a simple nominal design model. This is the model reduction process discussed in the previous chapters.

The following terminologies are used in this book.

Definition 9.1 Given the description of an uncertainty model set $\mathbf{\Pi}$ and a set of performance objectives, suppose $P \in \mathbf{\Pi}$ is the nominal design model and K is the resulting controller. Then the closed-loop feedback system is said to have

Nominal Stability (NS): if K internally stabilizes the nominal model P .

Robust Stability (RS): if K internally stabilizes every plant belong to $\mathbf{\Pi}$.

Nominal Performance (NP): if the performance objectives are satisfied for the nominal plant P .

Robust Performance (RP): if the performance objectives are satisfied for every plant belong to $\mathbf{\Pi}$.

The nominal stability and performance can be easily checked using various standard techniques. The conditions for which the robust stability and robust performance are satisfied under various assumptions on the uncertainty set $\mathbf{\Pi}$ will be considered in the following sections.

9.2 Small Gain Theorem

This section and the next section consider the stability test of a nominally stable system under unstructured perturbations. The basis for the robust stability criteria derived in the sequel is the so-called *small gain theorem*.

Consider the interconnected system shown in Figure 9.3 with $M(s)$ a stable $p \times q$ transfer matrix.

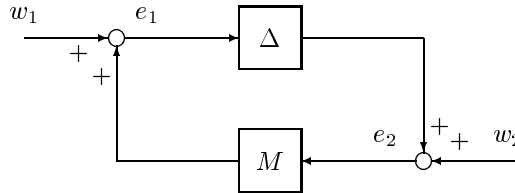


Figure 9.3: Small Gain Theorem

¹See, for example, Safonov [1980] and Zames [1966].

Theorem 9.1 (Small Gain Theorem) *Suppose $M \in \mathcal{RH}_\infty$. Then the interconnected system shown in Figure 9.3 is well-posed and internally stable for all $\Delta(s) \in \mathcal{RH}_\infty$ with*

- (a) $\|\Delta\|_\infty \leq 1/\gamma$ if and only if $\|M(s)\|_\infty < \gamma$;
- (b) $\|\Delta\|_\infty < 1/\gamma$ if and only if $\|M(s)\|_\infty \leq \gamma$.

Proof. We shall only prove part (a). The proof for part (b) is similar. Without loss of generality, assume $\gamma = 1$.

(Sufficiency) It is clear that $M(s)\Delta(s)$ is stable since both $M(s)$ and $\Delta(s)$ are stable. Thus by Theorem 5.7 (or Corollary 5.6) the closed-loop system is stable if $\det(I - M\Delta)$ has no zero in the closed right-half plane for all $\Delta \in \mathcal{RH}_\infty$ and $\|\Delta\|_\infty \leq 1$. Equivalently, the closed-loop system is stable if

$$\inf_{s \in \overline{\mathbb{C}}_+} \underline{\sigma}(I - M(s)\Delta(s)) \neq 0$$

for all $\Delta \in \mathcal{RH}_\infty$ and $\|\Delta\|_\infty \leq 1$. But this follows from

$$\inf_{s \in \overline{\mathbb{C}}_+} \underline{\sigma}(I - M(s)\Delta(s)) \geq 1 - \sup_{s \in \overline{\mathbb{C}}_+} \bar{\sigma}(M(s)\Delta(s)) = 1 - \|M(s)\Delta(s)\|_\infty \geq 1 - \|M(s)\|_\infty > 0.$$

(Necessity) This will be shown by contradiction. Suppose $\|M\|_\infty \geq 1$. We will show that there exists a $\Delta \in \mathcal{RH}_\infty$ with $\|\Delta\|_\infty \leq 1$ such that $\det(I - M(s)\Delta(s))$ has a zero on the imaginary axis, so the system is unstable. Suppose $\omega_0 \in \mathbb{R}_+ \cup \{\infty\}$ is such that $\bar{\sigma}(M(j\omega_0)) \geq 1$. Let $M(j\omega_0) = U(j\omega_0)\Sigma(j\omega_0)V^*(j\omega_0)$ be a singular value decomposition with

$$\begin{aligned} U(j\omega_0) &= \begin{bmatrix} u_1 & u_2 & \cdots & u_p \end{bmatrix} \\ V(j\omega_0) &= \begin{bmatrix} v_1 & v_2 & \cdots & v_q \end{bmatrix} \\ \Sigma(j\omega_0) &= \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \end{bmatrix} \end{aligned}$$

and $\|M\|_\infty = \bar{\sigma}(M(j\omega_0)) = \sigma_1$. To obtain a contradiction, it now suffices to construct a $\Delta \in \mathcal{RH}_\infty$ such that $\Delta(j\omega_0) = \frac{1}{\sigma_1}v_1u_1^*$ and $\|\Delta\|_\infty \leq 1$. Indeed, for such $\Delta(s)$,

$$\det(I - M(j\omega_0)\Delta(j\omega_0)) = \det(I - U\Sigma V^*v_1u_1^*/\sigma_1) = 1 - u_1^*U\Sigma V^*v_1/\sigma_1 = 0$$

and thus the closed-loop system is either not well-posed (if $\omega_0 = \infty$) or unstable (if $\omega_0 \in \mathbb{R}$). There are two different cases:

- (1) $\omega_0 = 0$ or ∞ : then U and V are real matrices. In this case, $\Delta(s)$ can be chosen as

$$\Delta = \frac{1}{\sigma_1}v_1u_1^* \in \mathbb{R}^{q \times p}.$$

(2) $0 < \omega_0 < \infty$: write u_1 and v_1 in the following form:

$$u_1 = \begin{bmatrix} u_{11}e^{j\theta_1} \\ u_{12}e^{j\theta_2} \\ \vdots \\ u_{1p}e^{j\theta_p} \end{bmatrix}, \quad v_1 = \begin{bmatrix} v_{11}e^{j\phi_1} \\ v_{12}e^{j\phi_2} \\ \vdots \\ v_{1q}e^{j\phi_q} \end{bmatrix}$$

where $u_{1i} \in \mathbb{R}$ and $v_{1j} \in \mathbb{R}$ are chosen so that $\theta_i, \phi_j \in [-\pi, 0)$ for all i, j .

Choose $\beta_i \geq 0$ and $\alpha_j \geq 0$ so that

$$\angle \left(\frac{\beta_i - j\omega_0}{\beta_i + j\omega_0} \right) = \theta_i, \quad \angle \left(\frac{\alpha_j - j\omega_0}{\alpha_j + j\omega_0} \right) = \phi_j$$

for $i = 1, 2, \dots, p$ and $j = 1, 2, \dots, q$. Let

$$\Delta(s) = \frac{1}{\sigma_1} \begin{bmatrix} v_{11} \frac{\alpha_1 - s}{\alpha_1 + s} \\ \vdots \\ v_{1q} \frac{\alpha_q - s}{\alpha_q + s} \end{bmatrix} \begin{bmatrix} u_{11} \frac{\beta_1 - s}{\beta_1 + s} & \cdots & u_{1p} \frac{\beta_p - s}{\beta_p + s} \end{bmatrix} \in \mathcal{RH}_\infty.$$

Then $\|\Delta\|_\infty = 1/\sigma_1 \leq 1$ and $\Delta(j\omega_0) = \frac{1}{\sigma_1} v_1 u_1^*$.

□

The theorem still holds even if Δ and M are infinite dimensional. This is summarized as the following corollary.

Corollary 9.2 *The following statements are equivalent:*

- (i) *The system is well-posed and internally stable for all $\Delta \in \mathcal{H}_\infty$ with $\|\Delta\|_\infty < 1/\gamma$;*
- (ii) *The system is well-posed and internally stable for all $\Delta \in \mathcal{RH}_\infty$ with $\|\Delta\|_\infty < 1/\gamma$;*
- (iii) *The system is well-posed and internally stable for all $\Delta \in \mathbb{C}^{q \times p}$ with $\|\Delta\| < 1/\gamma$;*
- (iv) $\|M\|_\infty \leq \gamma$.

Remark 9.1 It can be shown that the small gain condition is sufficient to guarantee internal stability even if Δ is a nonlinear and time varying “stable” operator with an appropriately defined stability notion, see Desoer and Vidyasagar [1975]. ♡

The following lemma shows that if $\|M\|_\infty > \gamma$, there exists a destabilizing Δ with $\|\Delta\|_\infty < 1/\gamma$ such that the closed-loop system has poles in the open right half plane. (This is stronger than what is given in the proof of Theorem 9.1.)

Lemma 9.3 *Suppose $M \in \mathcal{RH}_\infty$ and $\|M\|_\infty > \gamma$. Then there exists a $\sigma_0 > 0$ such that for any given $\sigma \in [0, \sigma_0]$ there exists a $\Delta \in \mathcal{RH}_\infty$ with $\|\Delta\|_\infty < 1/\gamma$ such that $\det(I - M(s)\Delta(s))$ has a zero on the axis $\operatorname{Re}(s) = \sigma$.*

Proof. Without loss of generality, assume $\gamma = 1$. Since $M \in \mathcal{RH}_\infty$ and $\|M\|_\infty > 1$, there is a sufficiently small $\sigma_0 > 0$ such that $\|M(\sigma_0 + s)\|_\infty := \sup_{\text{Re}(s) > 0} \|M(\sigma_0 + s)\| > 1$.

Now let $\sigma \in [0, \sigma_0]$. Then there exists a $0 < \omega_0 < \infty$ such that $\|M(\sigma + j\omega_0)\| > 1$.

Let $M(\sigma + j\omega_0) = U\Sigma V^*$ be a singular value decomposition with

$$U = \begin{bmatrix} u_1 & u_2 & \cdots & u_p \end{bmatrix}$$

$$V = \begin{bmatrix} v_1 & v_2 & \cdots & v_q \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \end{bmatrix}.$$

Write u_1 and v_1 in the following form:

$$u_1 = \begin{bmatrix} u_{11}e^{j\theta_1} \\ u_{12}e^{j\theta_2} \\ \vdots \\ u_{1p}e^{j\theta_p} \end{bmatrix}, \quad v_1 = \begin{bmatrix} v_{11}e^{j\phi_1} \\ v_{12}e^{j\phi_2} \\ \vdots \\ v_{1q}e^{j\phi_q} \end{bmatrix}$$

where $u_{1i} \in \mathbb{R}$ and $v_{1j} \in \mathbb{R}$ are chosen so that $\theta_i, \phi_j \in [-\pi, 0)$ for all i, j .

Choose $\beta_i \geq 0$ and $\alpha_j \geq 0$ so that

$$\angle \left(\frac{\beta_i - \sigma - j\omega_0}{\beta_i + \sigma + j\omega_0} \right) = \theta_i, \quad \angle \left(\frac{\alpha_j - \sigma - j\omega_0}{\alpha_j + \sigma + j\omega_0} \right) = \phi_j$$

for $i = 1, 2, \dots, p$ and $j = 1, 2, \dots, q$. Let

$$\Delta(s) = \frac{1}{\sigma_1} \begin{bmatrix} v_{11} \frac{\alpha_1 - s}{\alpha_1 + s} \\ \vdots \\ v_{1q} \frac{\alpha_q - s}{\alpha_q + s} \end{bmatrix} \begin{bmatrix} u_{11} \frac{\beta_1 - s}{\beta_1 + s} & \cdots & u_{1p} \frac{\beta_p - s}{\beta_p + s} \end{bmatrix} \in \mathcal{RH}_\infty.$$

Then $\|\Delta\|_\infty = 1/\sigma_1 < 1$ and $\det(I - M(\sigma + j\omega_0)\Delta(\sigma + j\omega_0)) = 0$. Hence $s = \sigma + j\omega_0$ is a zero for the transfer function $\det(I - M(s)\Delta(s))$. $\square$

The above lemma plays a key role in the necessity proofs of many robust stability tests in the sequel.

9.3 Stability under Stable Unstructured Uncertainties

The small gain theorem in the last section will be used here to derive robust stability tests under various assumptions of model uncertainties. The modeling error Δ will again

be assumed to be stable. (Most of the robust stability tests discussed in the sequel can be easily generalized to unstable Δ case with some mild assumptions on the number of unstable poles of the uncertain model, we encourage readers to fill in the details.) In addition, we assume that the modeling error Δ is suitably scaled with weighting functions W_1 and W_2 , i.e., the uncertainty can be represented as $W_1\Delta W_2$.

We shall consider the standard setup shown in Figure 9.4, where Π is the set of uncertain plants with $P \in \Pi$ as the nominal plant and with K as the internally stabilizing controller for P . The sensitivity and complementary sensitivity matrix functions are defined as usual as

$$S_o = (I + PK)^{-1}, \quad T_o = I - S_o$$

and

$$S_i = (I + KP)^{-1}, \quad T_i = I - S_i.$$

Recall that the closed-loop system is well-posed and internally stable if and only if

$$\begin{bmatrix} I & K \\ -\Pi & I \end{bmatrix}^{-1} = \begin{bmatrix} (I + K\Pi)^{-1} & -K(I + \Pi K)^{-1} \\ (I + \Pi K)^{-1}\Pi & (I + \Pi K)^{-1} \end{bmatrix} \in \mathcal{RH}_\infty.$$

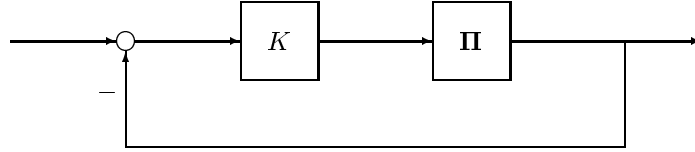


Figure 9.4: Unstructured Robust Stability Analysis

9.3.1 Additive Uncertainty

We assume that the model uncertainty can be represented by an additive perturbation:

$$\Pi = P + W_1\Delta W_2.$$

Theorem 9.4 *Let $\Pi = \{P + W_1\Delta W_2 : \Delta \in \mathcal{RH}_\infty\}$ and let K be a stabilizing controller for the nominal plant P . Then the closed-loop system is well-posed and internally stable for all $\|\Delta\|_\infty < 1$ if and only if $\|W_2KS_oW_1\|_\infty \leq 1$.*

Proof. Let $\Pi = P + W_1\Delta W_2$. Then

$$\begin{bmatrix} I & K \\ -\Pi & I \end{bmatrix}^{-1} = \begin{bmatrix} (I + KS_oW_1\Delta W_2)^{-1}S_i & -KS_o(I + W_1\Delta W_2KS_o)^{-1} \\ (I + S_oW_1\Delta W_2K)^{-1}S_o(P + W_1\Delta W_2) & S_o(I + W_1\Delta W_2KS_o)^{-1} \end{bmatrix}$$

is well-posed and internally stable if $(I + \Delta W_2KS_oW_1)^{-1} \in \mathcal{RH}_\infty$ since

$$\begin{aligned} \det(I + KS_oW_1\Delta W_2) &= \det(I + W_1\Delta W_2KS_o) = \det(I + S_oW_1\Delta W_2K) \\ &= \det(I + \Delta W_2KS_oW_1). \end{aligned}$$

But $(I + \Delta W_2 K S_o W_1)^{-1} \in \mathcal{RH}_\infty$ is guaranteed if $\|\Delta W_2 K S_o W_1\|_\infty < 1$. Hence $\|W_2 K S_o W_1\|_\infty \leq 1$ is sufficient for robust stability.

To show the necessity, note that robust stability implies that

$$K(I + \Pi K)^{-1} = K S_o (I + W_1 \Delta W_2 K S_o)^{-1} \in \mathcal{RH}_\infty$$

for all admissible Δ . This in turn implies that

$$\Delta W_2 K (I + \Pi K)^{-1} W_1 = I - (I + \Delta W_2 K S_o W_1)^{-1} \in \mathcal{RH}_\infty$$

for all admissible Δ . By small gain theorem, this is true for all $\Delta \in \mathcal{RH}_\infty$ with $\|\Delta\|_\infty < 1$ only if $\|W_2 K S_o W_1\|_\infty \leq 1$. $\square$

Similarly, it is easy to show that the closed-loop system is stable for all $\Delta \in \mathcal{RH}_\infty$ with $\|\Delta\|_\infty \leq 1$ if and only if $\|W_2 K S_o W_1\|_\infty < 1$.

9.3.2 Multiplicative Uncertainty

In this section, we assume that the system model is described by the following set of multiplicative perturbation

$$P_\Delta = (I + W_1 \Delta W_2)P$$

with $W_1, W_2, \Delta \in \mathcal{RH}_\infty$. Consider the feedback system shown in the Figure 9.5.

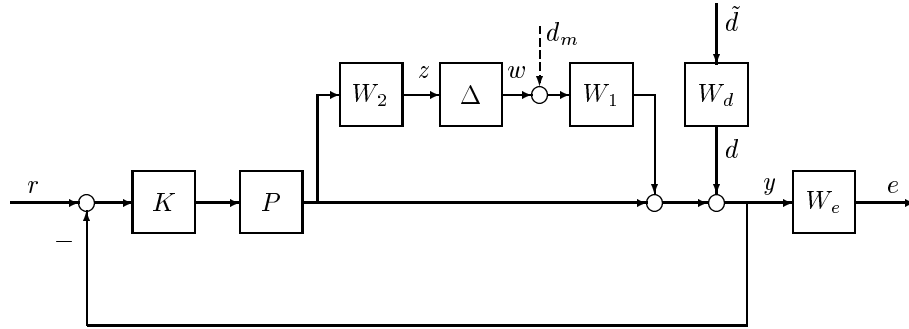


Figure 9.5: Output Multiplicative Perturbed Systems

Theorem 9.5 Let $\Pi = \{(I + W_1 \Delta W_2)P : \Delta \in \mathcal{RH}_\infty\}$ and let K be a stabilizing controller for the nominal plant P . Then

- (i) the closed-loop system is well-posed and internally stable for all $\Delta \in \mathcal{RH}_\infty$ with $\|\Delta\|_\infty < 1$ if and only if $\|W_2 T_o W_1\|_\infty \leq 1$.
- (ii) the closed-loop system is well-posed and internally stable for all $\Delta \in \mathcal{RH}_\infty$ with $\|\Delta\|_\infty \leq 1$ if $\|W_2 T_o W_1\|_\infty < 1$.

- (iii) the robust stability of the closed-loop system for all $\Delta \in \mathcal{RH}_\infty$ with $\|\Delta\|_\infty \leq 1$ does not necessarily imply $\|W_2 T_o W_1\|_\infty < 1$.
- (iv) the closed-loop system is well-posed and internally stable for all $\Delta \in \mathcal{RH}_\infty$ with $\|\Delta\|_\infty \leq 1$ only if $\|W_2 T_o W_1\|_\infty \leq 1$.
- (v) In addition, assume that neither P nor K has poles on the imaginary axis. Then the closed-loop system is well-posed and internally stable for all $\Delta \in \mathcal{RH}_\infty$ with $\|\Delta\|_\infty \leq 1$ if and only if $\|W_2 T_o W_1\|_\infty < 1$.

Proof. We shall first prove that the condition in (i) is necessary for robust stability. Suppose $\|W_2 T_o W_1\|_\infty > 1$. Then by Lemma 9.3, for any given sufficiently small $\sigma > 0$, there is a $\Delta \in \mathcal{RH}_\infty$ with $\|\Delta\|_\infty < 1$ such that $(I + \Delta W_2 T_o W_1)^{-1}$ has poles on the axis $\text{Re}(s) = \sigma$. This implies

$$(I + \Pi K)^{-1} = S_o(I + W_1 \Delta W_2 T_o)^{-1}$$

has poles on the axis $\text{Re}(s) = \sigma$ since σ can always be chosen so that the unstable poles are not cancelled by the zeros of S_o . Hence $\|W_2 T_o W_1\|_\infty \leq 1$ is necessary for robust stability. In fact, we have also proven part (iv). The sufficiency parts of (i), (ii), and (v) follow from the small gain theorem.

To show the necessity part of (v), suppose $\|W_2 T_o W_1\|_\infty = 1$. From the proof of the small gain theorem, there is a $\Delta \in \mathcal{RH}_\infty$ with $\|\Delta\|_\infty \leq 1$ such that $(I + \Delta W_2 T_o W_1)^{-1}$ has poles on the imaginary axis. This implies

$$(I + \Pi K)^{-1} = S_o(I + W_1 \Delta W_2 T_o)^{-1}$$

has poles on the imaginary axis since the imaginary axis poles of $(I + W_1 \Delta W_2 T_o)^{-1}$ are not cancelled by the zeros of S_o , which are the poles of P and K . Hence $\|W_2 T_o W_1\|_\infty < 1$ is necessary for robust stability.

The proof of part (iii) is given below by exhibiting an example with $\|W_2 T_o W_1\|_\infty = 1$ but there is no destabilizing $\Delta \in \mathcal{RH}_\infty$ with $\|\Delta\|_\infty \leq 1$. $\square$

Example 9.1 Let $P(s) = \frac{1}{s}$, $K(s) = 1$, and $W_1 = W_2 = 1$. It is easy to check that K stabilizes P . We have

$$T_o = \frac{1}{s+1}, \quad \|T_o\|_\infty = 1$$

and

$$(I + \Pi K)^{-1} = (I + K \Pi)^{-1} = K(I + \Pi K)^{-1} = \frac{s}{s+1} \frac{1}{1 + \frac{1}{s+1} \Delta}$$

$$(I + \Pi K)^{-1} \Pi = \frac{1}{s+1} \frac{1 + \Delta}{1 + \frac{1}{s+1} \Delta}.$$

Since $|T_o(s)| = \left| \frac{1}{s+1} \right| < 1$ for all $0 \neq s \in \mathbb{C}$ and $\operatorname{Re}(s) \geq 0$, $1 + \frac{1}{s+1}\Delta \neq 0$ for all $0 \neq s \in \mathbb{C}$, $\operatorname{Re}(s) \geq 0$, $\Delta \in \mathcal{RH}_\infty$ and $\|\Delta\| \leq 1$. The only point where $1 + \frac{1}{s+1}\Delta = 0$ in the closed right half plane is $s = 0$. Then $\Delta(0) = -1$. By assumption, Δ is analytic in a neighborhood of the origin since $\Delta \in \mathcal{RH}_\infty$. Hence, we can write

$$\Delta(s) = -1 + \sum_{i=1}^{\infty} a_i s^i, \quad a_i \in \mathbb{R}.$$

We now claim that $a_1 \geq 0$. Otherwise, $\left. \frac{d\Delta(s)}{ds} \right|_{s=0} = a_1 < 0$, along with $\Delta(0) = -1$, implies $\|\Delta\|_\infty > 1$. Hence $\Delta(s) = -1 + a_1 s + s^2 g(s)$ for some $g(s)$ and $a_1 \geq 0$ and

$$(I + \Pi K)^{-1} = (I + K\Pi)^{-1} = K(I + \Pi K)^{-1} = \frac{s}{s+1} \frac{1}{1 + \frac{1}{s+1}\Delta} = \frac{1}{1 + a_1 + sg}$$

$$(I + \Pi K)^{-1} \Pi = \frac{1}{s+1} \frac{1 + \Delta}{1 + \frac{1}{s+1}\Delta} = \frac{a_1 + sg}{1 + a_1 + sg}$$

both of which are bounded in the neighborhood of the origin. Hence there is no destabilizing $\Delta \in \mathcal{RH}_\infty$ with $\|\Delta\|_\infty \leq 1$. $\diamond$

The gap between the necessity and the sufficiency of the above robust stability conditions for the closed ball of uncertainty is only technique and will not be considered in the sequel. The reason for the existence of a such gap may be attributed to the fact that in the multiplicative perturbed case, the signals z , w , and d_m in Figure 9.5 are artificial and they are not physical signals. Indeed, $\Pi = (I + W_1 \Delta W_2)P$ is a single system, the internal stability of the closed-loop system does not necessarily imply the boundedness of the artificial signals z or w with respect to the artificial disturbance d_m . This is the case for the above example where $\Pi = (1 + \Delta)P = (a_1 s + s^2 g(s))/s = a_1 + sg(s)$ and the pole $s = 0$ is cancelled. This cancellation is artificial and is caused by the particular *model representation* (i.e., there is really no cancellation in the physical system.) Thus the closed-loop system is robustly stable although the transfer function from d_m to z is unstable.

9.3.3 Coprime Factor Uncertainty

As another example, consider a left coprime factor perturbed plant described in Figure 9.6.

Theorem 9.6 *Let*

$$\Pi = (\tilde{M} + \tilde{\Delta}_M)^{-1}(\tilde{N} + \tilde{\Delta}_N)$$

with $\tilde{M}, \tilde{N}, \tilde{\Delta}_M, \tilde{\Delta}_N \in \mathcal{RH}_\infty$. The transfer matrices $(\tilde{M}, \tilde{N})$ are assumed to be a stable left coprime factorization of P (i.e., $P = \tilde{M}^{-1}\tilde{N}$), and K internally stabilizes

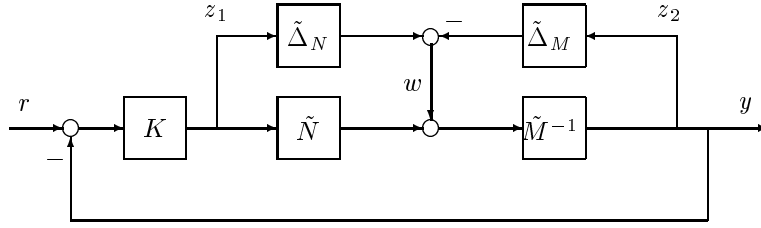


Figure 9.6: Left Coprime Factor Perturbed Systems

the nominal system P . Define $\Delta := \begin{bmatrix} \tilde{\Delta}_N & \tilde{\Delta}_M \end{bmatrix}$. Then the closed-loop system is well-posed and internally stable for all $\|\Delta\|_\infty < 1$ if and only if

$$\left\| \begin{bmatrix} K \\ I \end{bmatrix} (I + PK)^{-1} \tilde{M}^{-1} \right\|_\infty \leq 1.$$

Proof. Let $K = UV^{-1}$ be a right coprime factorization over $\mathcal{RH}_\infty$. By Lemma 5.10, the closed-loop system is internally stable if and only if

$$\left((\tilde{N} + \tilde{\Delta}_N)U + (\tilde{M} + \tilde{\Delta}_M)V \right)^{-1} \in \mathcal{RH}_\infty. \quad (9.3)$$

Since K stabilizes P , $(\tilde{N}U + \tilde{M}V)^{-1} \in \mathcal{RH}_\infty$. Hence (9.3) holds if and only if

$$\left(I + (\tilde{\Delta}_N U + \tilde{\Delta}_M V)(\tilde{N}U + \tilde{M}V)^{-1} \right)^{-1} \in \mathcal{RH}_\infty.$$

By the small gain theorem, the above is true for all $\|\Delta\|_\infty < 1$ if and only if

$$\left\| \begin{bmatrix} U \\ V \end{bmatrix} (\tilde{N}U + \tilde{M}V)^{-1} \right\|_\infty = \left\| \begin{bmatrix} K \\ I \end{bmatrix} (I + PK)^{-1} \tilde{M}^{-1} \right\|_\infty \leq 1.$$

□

Similarly, one can show that the closed-loop system is well-posed and internally stable for all $\|\Delta\|_\infty \leq 1$ if and only if

$$\left\| \begin{bmatrix} K \\ I \end{bmatrix} (I + PK)^{-1} \tilde{M}^{-1} \right\|_\infty < 1.$$

9.3.4 Unstructured Robust Stability Tests

Table 9.1 summaries robust stability tests on the plant uncertainties under various assumptions. All of the tests pertain to the standard setup shown in Figure 9.4, where Π is the set of uncertain plants with $P \in \Pi$ as the nominal plant and with K as the internally stabilizing controller of P .

Table 9.1 should be interpreted as

UNSTRUCTURED ANALYSIS THEOREM

Given **NS & Perturbed Model Sets**
 Then **Closed-Loop Robust Stability**
 if and only if **Robust Stability Tests**

The table also indicates representative types of physical uncertainties which can be usefully represented by cone bounded perturbations inserted at appropriate locations. For example, the representation $P_\Delta = (I + W_1 \Delta W_2)P$ in the first row is useful for output errors at high frequencies (HF), covering such things as unmodeled high frequency dynamics of sensors or plant, including diffusion processes, transport lags, electro-mechanical resonances, etc. The representation $P_\Delta = P(I + W_1 \Delta W_2)$ in the second row covers similar types of errors at the inputs. Both cases should be contrasted with the third and the fourth rows which treat $P(I + W_1 \Delta W_2)^{-1}$ and $(I + W_1 \Delta W_2)^{-1}P$. These representations are more useful for variations in modeled dynamics, such as low frequency (LF) errors produced by parameter variations with operating conditions, with aging, or across production copies of the same plant. Discussion of still other cases is left to the table.

Note from the table that the stability requirements on Δ do not limit our ability to represent variations in either the number or locations of rhp singularities as can be seen from some simple examples.

Example 9.2 Suppose an uncertain system with changing numbers of right-half plane poles is described by

$$P_\Delta = \left\{ \frac{1}{s - \delta} : \delta \in \mathbb{R}, |\delta| \leq 1 \right\}.$$

Then $P_1 = \frac{1}{s-1} \in P_\Delta$ has one right-half plane pole and $P_2 = \frac{1}{s+1} \in P_\Delta$ has no right-half plane pole. Nevertheless, the set of P_Δ can be covered by a set of feedback uncertain plants:

$$P_\Delta \subset \Pi := \{P(1 + \delta P)^{-1} : \delta \in \mathcal{RH}_\infty, \|\delta\|_\infty \leq 1\}$$

with $P = \frac{1}{s}$. ◇

Example 9.3 As another example, consider the following set of plants:

$$P_\Delta = \frac{s + 1 + \alpha}{(s + 1)(s + 2)}, |\alpha| \leq 2.$$

$W_1 \in \mathcal{RH}_\infty \quad W_2 \in \mathcal{RH}_\infty \quad \Delta \in \mathcal{RH}_\infty \quad \ \Delta\ _\infty < 1$		
Perturbed Model Sets Π	Representative Types of Uncertainty Characterized	Robust Stability Tests
$(I + W_1 \Delta W_2)P$	output (sensor) errors neglected HF dynamics changing # of rhp zeros	$\ W_2 T_o W_1\ _\infty \leq 1$
$P(I + W_1 \Delta W_2)$	input (actuators) errors neglected HF dynamics changing # of rhp zeros	$\ W_2 T_i W_1\ _\infty \leq 1$
$(I + W_1 \Delta W_2)^{-1}P$	LF parameter errors changing # of rhp poles	$\ W_2 S_o W_1\ _\infty \leq 1$
$P(I + W_1 \Delta W_2)^{-1}$	LF parameter errors changing # of rhp poles	$\ W_2 S_i W_1\ _\infty \leq 1$
$P + W_1 \Delta W_2$	additive plant errors neglected HF dynamics uncertain rhp zeros	$\ W_2 K S_o W_1\ _\infty \leq 1$
$P(I + W_1 \Delta W_2 P)^{-1}$	LF parameter errors uncertain rhp poles	$\ W_2 S_o P W_1\ _\infty \leq 1$
$(\tilde{M} + \tilde{\Delta}_M)^{-1}(\tilde{N} + \tilde{\Delta}_N)$ $P = \tilde{M}^{-1}\tilde{N}$ $\Delta = \begin{bmatrix} \tilde{\Delta}_N & \tilde{\Delta}_M \end{bmatrix}$	LF parameter errors neglected HF dynamics uncertain rhp poles & zeros	$\left\ \begin{bmatrix} K \\ I \end{bmatrix} S_o \tilde{M}^{-1} \right\ _\infty \leq 1$
$(N + \Delta_N)(M + \Delta_M)^{-1}$ $P = N M^{-1}$ $\Delta = \begin{bmatrix} \Delta_N \\ \Delta_M \end{bmatrix}$	LF parameter errors neglected HF dynamics uncertain rhp poles & zeros	$\ M^{-1} S_i [K \ I]\ _\infty \leq 1$

Table 9.1: Unstructured Robust Stability Tests (# stands for ‘the number’.)

This set of plants have *changing numbers of right-half plane zeros* since the plant has no right-half plane zero when $\alpha = 0$ and has one right-half plane zero when $\alpha = -2$. The uncertain plant can be covered by a set of multiplicative perturbed plants:

$$P_{\Delta} \subset \Pi := \left\{ \frac{1}{s+2} \left(1 + \frac{2\delta}{s+1} \right), \delta \in \mathcal{RH}_{\infty}, \|\delta\|_{\infty} \leq 1 \right\}.$$

It should be noted that this covering can be quite conservative. $\diamond$

9.3.5 Equivalence: Robust Stability vs. Nominal Performance

A robust stability problem can be viewed as another nominal performance problem. For example, the output multiplicative perturbed robust stability problem can be treated as a sensor noise rejection problem shown in Figure 9.7 and vice versa. It is clear that the system with output multiplicative uncertainty as shown in Figure 9.5 is robustly stable for $\|\Delta\|_{\infty} < 1$ iff the $\mathcal{H}_{\infty}$ norm of the transfer function from w to z , T_{zw} , is no greater than 1. Since $T_{zw} = T_e \tilde{n}$, hence $\|T_{zw}\|_{\infty} \leq 1$ iff $\sup_{\|\tilde{n}\|_2 \leq 1} \|e\|_2 = \|W_2 T_o W_1\|_{\infty} \leq 1$.

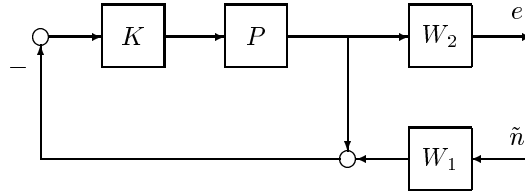


Figure 9.7: Equivalence Between Robust Stability With Output Multiplicative Uncertainty and Nominal Performance With Sensor Noise Rejection

There is, in fact, a much more general result along this line of equivalence: any robust stability problem (with open ball of uncertainty) can be regarded as an equivalent performance problem. This will be considered in Chapter 11.

9.4 Unstructured Robust Performance

Consider the perturbed system shown in Figure 9.8 with the set of perturbed models described by a set Π . Suppose the weighting matrices $W_d, W_e \in \mathcal{RH}_{\infty}$ and the performance criterion is to keep the error e as small as possible in some sense for all possible models belong to the set Π . In general, the set Π can be either parameterized set or unstructured set such as those described in Table 9.1. The performance specifications are usually specified in terms of the magnitude of each component e in time domain, i.e., $\mathcal{L}_{\infty}$ norm, with respect to bounded disturbances, or alternatively and more conveniently some requirements on the closed-loop frequency response of the transfer matrix between $\tilde{d}$ and e , say, integral of square error or the magnitude of the steady state error with

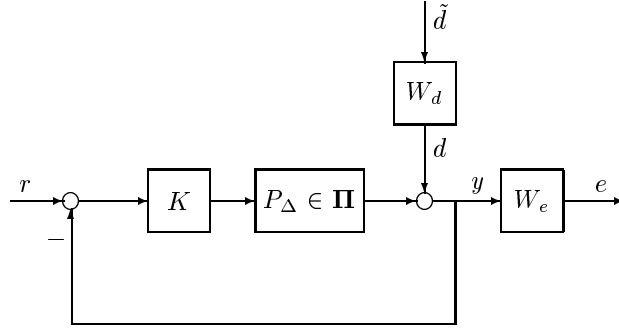


Figure 9.8: Diagram for Robust Performance Analysis

respect to sinusoidal disturbances. The former design criterion leads to the so-called $\mathcal{L}_1$ -optimal control framework and the latter leads to $\mathcal{H}_2$ and $\mathcal{H}_\infty$ design frameworks, respectively. In this section, we will focus primarily on the $\mathcal{H}_2$ and $\mathcal{H}_\infty$ performance objectives with unstructured model uncertainty descriptions. The performance under structured uncertainty will be considered in Chapter 11.

9.4.1 Robust $\mathcal{H}_2$ Performance

Although nominal $\mathcal{H}_2$ performance analysis is straightforward and involves only the computation of $\mathcal{H}_2$ norms, the $\mathcal{H}_2$ performance analysis with $\mathcal{H}_\infty$ norm bounded model uncertainty is much harder and little studied. Nevertheless, this problem is sensible and important for the reason that performance is sometimes more appropriately specified in terms of the $\mathcal{H}_2$ norm than the $\mathcal{H}_\infty$ norm. On the other hand, model uncertainties are more conveniently described using the $\mathcal{H}_\infty$ norm bounded sets and arise naturally in identification processes.

Let T_{ed}^Δ denote the transfer matrix between $\tilde{d}$ and e , then

$$T_{ed}^\Delta = W_e(I + P_\Delta K)^{-1}W_d, \quad P_\Delta \in \Pi. \quad (9.4)$$

And the robust $\mathcal{H}_2$ performance analysis problem can be stated as finding

$$\sup_{P_\Delta \in \Pi} \|T_{ed}^\Delta\|_2.$$

To simplify our analysis, consider a scalar system with $W_d = 1$, $W_e = w_s$, $P = p$, and assume that the model is given by a multiplicative uncertainty set

$$\Pi = \{(1 + w_t \delta)p : \delta \in \mathcal{RH}_\infty, \|\delta\|_\infty < 1\}.$$

Assume further that the system is robustly stabilized by a controller k . Then

$$\sup_{P_\Delta \in \Pi} \|T_{ed}^\Delta\|_2 = \sup_{\|\delta\|_\infty < 1} \left\| \frac{w_s \varepsilon}{1 + \tau w_t \delta} \right\|_2 = \left\| \frac{w_s \varepsilon}{1 - |\tau w_t|} \right\|_2$$

where $\varepsilon = (1 + pk)^{-1}$ and $\tau = 1 - \varepsilon$.

The exact analysis for the matrix case is harder to determine although some upper bounds can be derived as we shall do for the $\mathcal{H}_\infty$ case below. However, the upper bounds do not seem to be insightful in the $\mathcal{H}_2$ setting, and, therefore, are omitted. It should be pointed out that synthesis for robust $\mathcal{H}_2$ performance is much harder even in the scalar case although synthesis for nominal performance is relatively easy and will be considered in Chapter 14.

9.4.2 Robust $\mathcal{H}_\infty$ Performance with Output Multiplicative Uncertainty

Suppose the performance criterion is to keep the energy of error e as small as possible, i.e.,

$$\sup_{\|d\|_2 \leq 1} \|e\|_2 \leq \epsilon$$

for some small ϵ . By scaling the error e (i.e., by properly selecting W_e) we can assume without loss of generality that $\epsilon = 1$. Then the robust performance criterion in this case can be described as requiring that the closed-loop system be robustly stable and that

$$\|T_{ed}^\Delta\|_\infty \leq 1, \quad \forall P_\Delta \in \Pi. \quad (9.5)$$

More specifically, an output multiplicatively perturbed system will be analyzed first. The analysis for other classes of models can be done analogously. The perturbed model can be described as

$$\Pi := \{(I + W_1 \Delta W_2)P : \Delta \in \mathcal{RH}_\infty, \|\Delta\|_\infty < 1\} \quad (9.6)$$

with $W_1, W_2 \in \mathcal{RH}_\infty$. The explicit system diagram is as shown in Figure 9.5. For this class of models, we have

$$T_{ed}^\Delta = W_e S_o (I + W_1 \Delta W_2 T_o)^{-1} W_d,$$

and the robust performance is satisfied iff

$$\|W_2 T_o W_1\|_\infty \leq 1$$

and

$$\|T_{ed}^\Delta\|_\infty \leq 1, \quad \forall \Delta \in \mathcal{RH}_\infty, \|\Delta\|_\infty < 1.$$

The exact analysis for this robust performance problem is not trivial and will be given in Chapter 11. However, some sufficient conditions are relatively easy to obtain by bounding these two inequalities, and they may shed some light on the nature of these problems. It will be assumed throughout that the controller K internally stabilizes the nominal plant P .

Theorem 9.7 Suppose $P_\Delta \in \{(I + W_1\Delta W_2)P : \Delta \in \mathcal{RH}_\infty, \|\Delta\|_\infty < 1\}$ and K internally stabilizes P . Then the system robust performance is guaranteed if either one of the following conditions is satisfied

(i) for each frequency ω

$$\bar{\sigma}(W_d)\bar{\sigma}(W_e S_o) + \bar{\sigma}(W_1)\bar{\sigma}(W_2 T_o) \leq 1; \quad (9.7)$$

(ii) for each frequency ω

$$\kappa(W_1^{-1}W_d)\bar{\sigma}(W_e S_o W_d) + \bar{\sigma}(W_2 T_o W_1) \leq 1 \quad (9.8)$$

where W_1 and W_d are assumed to be invertible and $\kappa(W_1^{-1}W_d)$ is the condition number.

Proof. It is obvious that both condition (9.7) and condition (9.8) guarantee that $\|W_2 T_o W_1\|_\infty \leq 1$. So it is sufficient to show that $\|T_{ed}^\Delta\|_\infty \leq 1, \forall \Delta \in \mathcal{RH}_\infty, \|\Delta\|_\infty < 1$. Now for any frequency ω , it is easy to see that

$$\begin{aligned} \bar{\sigma}(T_{ed}^\Delta) &\leq \bar{\sigma}(W_e S_o)\bar{\sigma}[(I + W_1\Delta W_2 T_o)^{-1}]\bar{\sigma}(W_d) \\ &= \frac{\bar{\sigma}(W_e S_o)\bar{\sigma}(W_d)}{\underline{\sigma}(I + W_1\Delta W_2 T_o)} \\ &\leq \frac{\bar{\sigma}(W_e S_o)\bar{\sigma}(W_d)}{1 - \bar{\sigma}(W_1\Delta W_2 T_o)} \\ &\leq \frac{\bar{\sigma}(W_e S_o)\bar{\sigma}(W_d)}{1 - \bar{\sigma}(W_1)\bar{\sigma}(W_2 T_o)\bar{\sigma}(\Delta)}. \end{aligned}$$

Hence condition (9.7) guarantees $\bar{\sigma}(T_{ed}^\Delta) \leq 1$ for all $\Delta \in \mathcal{RH}_\infty$ with $\|\Delta\|_\infty < 1$ at all frequencies.

Similarly, suppose W_1 and W_d are invertible; write

$$T_{ed}^\Delta = W_e S_o W_d (W_1^{-1}W_d)^{-1} (I + \Delta W_2 T_o W_1)^{-1} (W_1^{-1}W_d),$$

and then

$$\bar{\sigma}(T_{ed}^\Delta) \leq \frac{\bar{\sigma}(W_e S_o W_d)\kappa(W_1^{-1}W_d)}{1 - \bar{\sigma}(W_2 T_o W_1)\bar{\sigma}(\Delta)}.$$

Hence by condition (9.8), $\bar{\sigma}(T_{ed}^\Delta) \leq 1$ is guaranteed for all $\Delta \in \mathcal{RH}_\infty$ with $\|\Delta\|_\infty < 1$ at all frequencies. $\square$

Remark 9.2 It is not hard to show that either one of the conditions in the theorem is also necessary for scalar valued systems.

Remark 9.3 Suppose $\kappa(W_1^{-1}W_d) \approx 1$ (weighting matrices satisfying this condition are usually called round weights). This is particularly the case if $W_1 = w_1(s)I$ and $W_d = w_d(s)I$. Recall that $\bar{\sigma}(W_e S_o W_d) \leq 1$ is the necessary and sufficient condition for nominal performance and that $\bar{\sigma}(W_2 T_o W_1) \leq 1$ is the necessary and sufficient condition for robust stability. Hence it is clear that in this case nominal performance plus robust stability almost guarantees robust performance, i.e., $\text{NP} + \text{RS} \approx \text{RP}$. However, this is not necessarily true for other types of uncertainties as will be shown later. ♡

Remark 9.4 Note that in light of the equivalence relation between the robust stability and nominal performance, it is reasonable to conjecture that the above robust performance problem is equivalent to the robust stability problem in Figure 9.4 with the uncertainty model set given by

$$\Pi := (I + W_d \Delta_e W_e)^{-1} (I + W_1 \Delta W_2) P$$

and $\|\Delta_e\|_\infty < 1$, $\|\Delta\|_\infty < 1$, as shown in Figure 9.9. This conjecture is indeed true; however, the equivalent model uncertainty is structured, and the exact stability analysis for such systems is not trivial and will be studied in Chapter 11. ♡

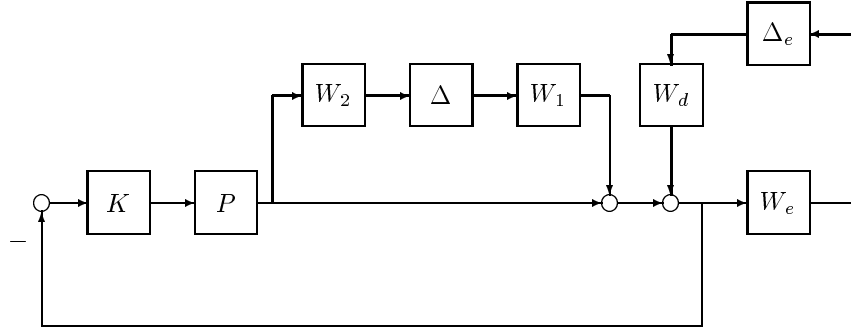


Figure 9.9: Robust Performance with Unstructured Uncertainty vs. Robust Stability with Structured Uncertainty

Remark 9.5 Note that if W_1 and W_d are invertible, then T_{ed}^Δ can also be written as

$$T_{ed}^\Delta = W_e S_o W_d [I + (W_1^{-1} W_d)^{-1} \Delta W_2 T_o W_1 (W_1^{-1} W_d)]^{-1}.$$

So another alternative sufficient condition for robust performance can be obtained as

$$\bar{\sigma}(W_e S_o W_d) + \kappa(W_1^{-1} W_d) \bar{\sigma}(W_2 T_o W_1) \leq 1.$$

A similar situation also occurs in the skewed case below. We will not repeat all these variations. ♡

9.4.3 Skewed Specifications and Plant Condition Number

We now consider the system with skewed specifications, i.e., the uncertainty and performance are not measured at the same location. For instance, the system performance is still measured in terms of output sensitivity, but the uncertainty model is in input multiplicative form:

$$\Pi := \{P(I + W_1\Delta W_2) : \Delta \in \mathcal{RH}_\infty, \|\Delta\|_\infty < 1\}.$$

For systems described by this class of models, see Figure 9.8, the robust stability condition becomes

$$\|W_2 T_i W_1\|_\infty \leq 1,$$

and the nominal performance condition becomes

$$\|W_e S_o W_d\|_\infty \leq 1.$$

To consider the robust performance, let $\tilde{T}_{ed}^\Delta$ denote the transfer matrix from $\tilde{d}$ to e . Then

$$\begin{aligned} \tilde{T}_{ed}^\Delta &= W_e S_o (I + P W_1 \Delta W_2 K S_o)^{-1} W_d \\ &= W_e S_o W_d [I + (W_d^{-1} P W_1) \Delta (W_2 T_i W_1) (W_d^{-1} P W_1)^{-1}]^{-1}. \end{aligned}$$

The last equality follows if W_1 , W_d , and P are invertible and, if W_2 is invertible, can also be written as

$$\tilde{T}_{ed}^\Delta = W_e S_o W_d (W_1^{-1} W_d)^{-1} [I + (W_1^{-1} P W_1) \Delta (W_2 P^{-1} W_2^{-1}) (W_2 T_o W_1)]^{-1} (W_1^{-1} W_d).$$

Then the following results follow easily.

Theorem 9.8 Suppose $P_\Delta \in \Pi = \{P(I + W_1\Delta W_2) : \Delta \in \mathcal{RH}_\infty, \|\Delta\|_\infty < 1\}$ and K internally stabilizes P . Assume that P, W_1, W_2 , and W_d are square and invertible. Then the system robust performance is guaranteed if either one of the following conditions is satisfied

(i) for each frequency ω

$$\overline{\sigma}(W_e S_o W_d) + \kappa(W_d^{-1} P W_1) \overline{\sigma}(W_2 T_i W_1) \leq 1; \quad (9.9)$$

(ii) for each frequency ω

$$\kappa(W_1^{-1} W_d) \overline{\sigma}(W_e S_o W_d) + \overline{\sigma}(W_1^{-1} P W_1) \overline{\sigma}(W_2 P^{-1} W_2^{-1}) \overline{\sigma}(W_2 T_o W_1) \leq 1. \quad (9.10)$$

Remark 9.6 If the appropriate invertibility conditions are not satisfied, then an alternative sufficient condition for robust performance can be given by

$$\overline{\sigma}(W_d) \overline{\sigma}(W_e S_o) + \overline{\sigma}(P W_1) \overline{\sigma}(W_2 K S_o) \leq 1.$$

Similar to the previous case, there are many different variations of sufficient conditions although (9.10) may be the most useful one. $\heartsuit$

Remark 9.7 It is important to note that in this case, the robust stability condition is given in terms of $L_i = KP$ while the nominal performance condition is given in terms of $L_o = PK$. These classes of problems are called *skewed problems* or problems with skewed specifications.² Since, in general, $PK \neq KP$, the robust stability margin or tolerances for uncertainties at the plant input and output are generally not the same. $\heartsuit$

Remark 9.8 It is also noted that the robust performance condition is related to the condition number of the weighted nominal model. So in general if the weighted nominal model is ill-conditioned at the range of critical frequencies, then the robust performance condition may be far more restrictive than the robust stability condition and the nominal performance condition together. For simplicity, assume $W_1 = I$, $W_d = I$ and $W_2 = w_t I$ where $w_t \in \mathcal{RH}_\infty$ is a scalar function. Further, P is assumed to be invertible. Then robust performance condition (9.10) can be written as

$$\bar{\sigma}(W_e S_o) + \kappa(P) \bar{\sigma}(w_t T_o) \leq 1, \forall \omega.$$

Comparing these conditions with those obtained for non-skewed problems shows that the condition related to robust stability is scaled by the condition number of the plant³. Since $\kappa(P) \geq 1$, it is clear that the skewed specifications are much harder to satisfy if the plant is not well conditioned. This problem will be discussed in more detail in section 11.3.3 of Chapter 11. $\heartsuit$

Remark 9.9 Suppose K is invertible, then $\tilde{T}_{ed}^\Delta$ can be written as

$$\tilde{T}_{ed}^\Delta = W_e K^{-1} (I + T_i W_1 \Delta W_2)^{-1} S_i K W_d.$$

Assume further that $W_e = I$, $W_d = w_s I$, $W_2 = I$ where $w_s \in \mathcal{RH}_\infty$ is a scalar function. Then a sufficient condition for robust performance is given by

$$\kappa(K) \bar{\sigma}(S_i w_s) + \bar{\sigma}(T_i W_1) \leq 1, \forall \omega,$$

with $\kappa(K) := \bar{\sigma}(K) \bar{\sigma}(K^{-1})$. This is equivalent to treating the input multiplicative plant uncertainty as the output multiplicative controller uncertainty. $\heartsuit$

These skewed specifications also create problems for MIMO loop shaping design which has been discussed briefly in Chapter 5. The idea of loop shaping design is based on the fact that robust performance is guaranteed by designing a controller with a sufficient nominal performance margin and a sufficient robust stability margin. For example, if $\kappa(W_1^{-1} W_d) \approx 1$, the output multiplicative perturbed robust performance is guaranteed by designing a controller with twice the required nominal performance and robust stability margins.

²See Stein and Doyle [1991].

³Alternative condition can be derived so that the condition related to nominal performance is scaled by the condition number.

The fact that the condition number appeared in the robust performance test for skewed problems can be given another interpretation by considering two sets of plants Π_1 and Π_2 as shown below.

$$\begin{aligned}\Pi_1 &:= \{P(I + w_t \Delta) : \Delta \in \mathcal{RH}_\infty, \|\Delta\|_\infty < 1\} \\ \Pi_2 &:= \{(I + \tilde{w}_t \Delta)P : \Delta \in \mathcal{RH}_\infty, \|\Delta\|_\infty < 1\}.\end{aligned}$$

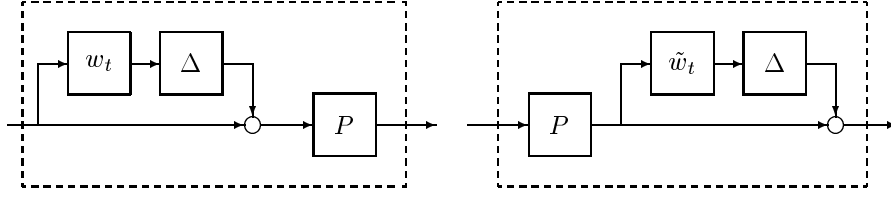


Figure 9.10: Converting Input Uncertainty to Output Uncertainty

Assume that P is invertible, then

$$\Pi_2 \supseteq \Pi_1 \quad \text{if} \quad |\tilde{w}_t| \geq |w_t| \kappa(P) \quad \forall \omega$$

since $P(I + w_t \Delta) = (I + w_t P \Delta P^{-1})P$.

The condition number of a transfer matrix can be very high at high frequency which may significantly limit the achievable performance. The example below, taken from the textbook [Franklin, Powell, and Workman, 1990], shows that the condition number shown in Figure 9.11 may increase with the frequency:

$$P(s) = \left[\begin{array}{ccc|cc} -0.2 & 0.1 & 1 & 0 & 1 \\ -0.05 & 0 & 0 & 0 & 0.7 \\ 0 & 0 & -1 & 1 & 0 \\ \hline 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{array} \right] = \frac{1}{a(s)} \begin{bmatrix} s & (s+1)(s+0.07) \\ -0.05 & 0.7(s+1)(s+0.13) \end{bmatrix}$$

where $a(s) = (s+1)(s+0.1707)(s+0.02929)$.

9.5 Gain Margin and Phase Margin

In this section, we show that the gain margin and phase margin defined in classical control theory may not be sufficient indicators of a system's robustness. Let $L(s)$ be a scalar transfer function and consider a unit feedback system such as the one shown in the following diagram:

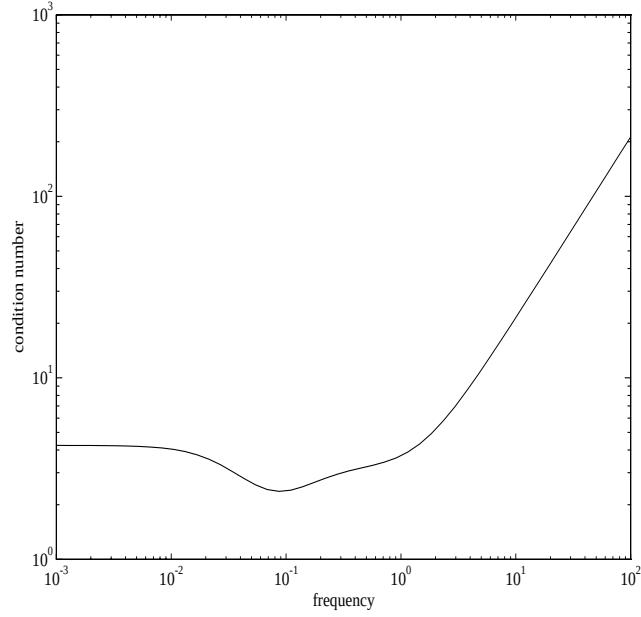
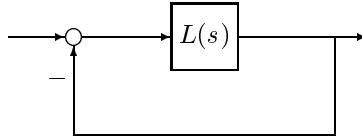


Figure 9.11: Condition Number $\kappa(\omega) = \bar{\sigma}(P(j\omega))/\underline{\sigma}(P(j\omega))$



Suppose that the above closed-loop feedback system with $L(s) = L_0(s)$ is stable. Then the system is said to have

Gain Margin $k_{\min}$ and $k_{\max}$ if the closed-loop system is stable for all $L(s) = kL_0(s)$ with $k_{\min} < k < k_{\max}$ but unstable for $L(s) = k_{\max}L_0(s)$ and for $L(s) = k_{\min}L_0(s)$ where $0 \leq k_{\min} \leq 1$ and $k_{\max} \geq 1$.

Phase Margin $\phi_{\min}$ and $\phi_{\max}$ if the closed-loop system is stable for all $L(s) = e^{-j\phi}L_0(s)$ with $\phi_{\min} < \phi < \phi_{\max}$ but unstable for $L(s) = e^{-j\phi_{\max}}L_0(s)$ and for $L(s) = e^{-j\phi_{\min}}L_0(s)$ where $-\pi \leq \phi_{\min} \leq 0$ and $0 \leq \phi_{\max} \leq \pi$.

These margins can be easily read from the open-loop system Nyquist diagram as shown in Figure 9.12 where $k_{\max}$ and $k_{\min}$ represent how much the loop gain can be increased

and decreased, respectively, without causing instability. Similarly $\phi_{\max}$ and $\phi_{\min}$ represent how much loop phase lag and lead can be tolerated without causing instability.

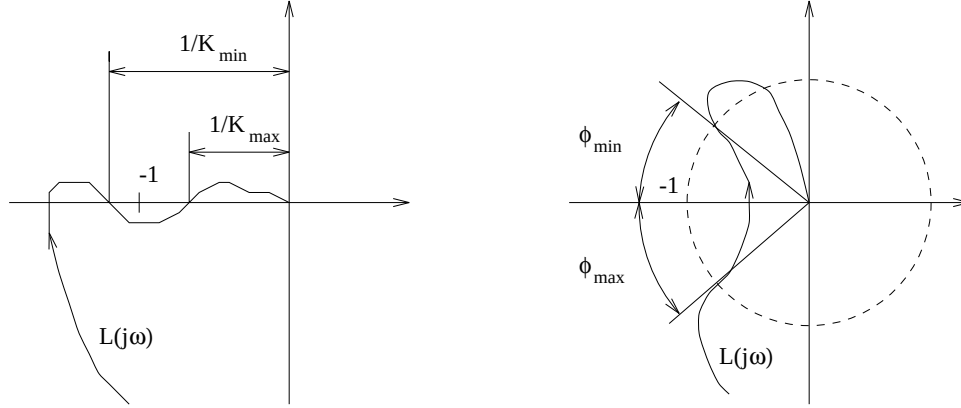


Figure 9.12: Gain Margin and Phase Margin of A Scalar System

However, gain margin or phase margin alone may not be a sufficient indicator of a system's robustness. To be more specific, consider a simple dynamical system

$$P = \frac{a-s}{as-1}, \quad a > 1$$

with a stabilizing controller K . Now let $L = PK$ and consider a controller

$$K = \frac{b+s}{bs+1}, \quad b > 0.$$

It is easy to show that the closed-loop system is stable for any

$$\frac{1}{a} < b < a.$$

To compute the stability margins, consider three cases:

- (i) $b = 1$: in this case, $K = 1$ and the stability margins can be easily shown to be

$$k_{\min} = \frac{1}{a}, \quad k_{\max} = a, \quad \phi_{\min} = -\pi, \quad \phi_{\max} = \sin^{-1} \left(\frac{a^2 - 1}{a^2 + 1} \right) =: \theta.$$

It is easy to see that both gain margin and phase margin are very large for large a .

- (ii) $\frac{1}{a} < b < a$ and $b \rightarrow a$: in this case

$$k_{\min} = \frac{1}{ab} \rightarrow \frac{1}{a^2}, \quad k_{\max} = ab \rightarrow a^2, \quad \phi_{\min} = -\pi, \quad \phi_{\max} \rightarrow 0$$

i.e., very large gain margin but arbitrarily small phase margin.

(iii) $\frac{1}{a} < b < a$ and $b \rightarrow \frac{1}{a}$: in this case

$$k_{\min} = \frac{1}{ab} \rightarrow 1, \quad k_{\max} = ab \rightarrow 1, \quad \phi_{\min} = -\pi, \quad \phi_{\max} \rightarrow 2\theta$$

i.e., very large phase margin but arbitrarily small gain margin.

The open-loop frequency response of the system is shown in Figure 9.13 for $a = 2$ and $b = 1, b = 1.9$ and $b = 0.55$, respectively.

Sometimes, gain margin and phase margin together may still not be enough to indicate the true robustness of a system. For example, it is possible to construct a (complicated) controller such that

$$k_{\min} < \frac{1}{a}, \quad k_{\max} > a, \quad \phi_{\min} = -\pi, \quad \phi_{\max} > \theta$$

but the Nyquist plot is arbitrarily close to $(-1, 0)$. Such a controller is complicated to construct; however, the following controller should give the reader a good idea of its construction:

$$K_{\text{bad}} = \frac{s + 3.3}{3.3s + 1} \frac{s + 0.55}{0.55s + 1} \frac{1.7s^2 + 1.5s + 1}{s^2 + 1.5s + 1.7}.$$

The open-loop frequency response of the system with this controller is also shown in Figure 9.13 by the dotted line. It is easy to see that the system has at least the same gain margin and phase margin as the system with controller $K = 1$, but the Nyquist plot is closer to $(-1, 0)$. Therefore this system is less robust with respect to the simultaneous gain and phase perturbations. The problem is that the gain margin and phase margin do not give the correct indication of the system's robustness when the gain and phase are varying simultaneously.

9.6 Deficiency of Classical Control for MIMO Systems

In this section, we show through an example that the classical control theory may not be reliable when it is applied to MIMO system design.

Consider a symmetric spinning body with torque inputs, T_1 and T_2 , along two orthogonal transverse axes, x and y , as shown in Figure 9.14. Assume that the angular velocity of the spinning body with respect to the z axis is constant, Ω . Assume further that the inertia of the spinning body with respect to the x, y , and z axes are $I_1, I_2 = I_1$, and I_3 , respectively. Denote by ω_1 and ω_2 the angular velocities of the body with respect to the x and y axes, respectively. Then the Euler's equation of the spinning body is given by

$$\begin{aligned} I_1 \dot{\omega}_1 - \omega_2 \Omega (I_1 - I_3) &= T_1 \\ I_1 \dot{\omega}_2 - \omega_1 \Omega (I_3 - I_1) &= T_2. \end{aligned}$$

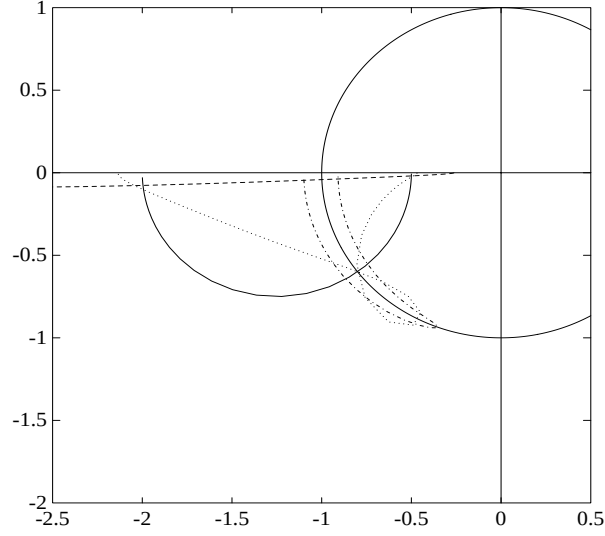


Figure 9.13: Nyquist Plots of L with $a = 2$ and $b = 1$ (solid), $b = 1.9$ (dashed), $b = 0.55$ (dashdot) and with K_{bad} (dotted)

Define

$$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} := \begin{bmatrix} T_1/I_1 \\ T_2/I_1 \end{bmatrix}, \quad a := (1 - I_3/I_1)\Omega.$$

Then the system dynamical equations can be written as

$$\begin{bmatrix} \dot{\omega}_1 \\ \dot{\omega}_2 \end{bmatrix} = \begin{bmatrix} 0 & a \\ -a & 0 \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \end{bmatrix} + \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}.$$

Now suppose that the angular rates ω_1 and ω_2 are measured in scaled and rotated coordinates:

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \frac{1}{\cos \theta} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \end{bmatrix} = \begin{bmatrix} 1 & a \\ -a & 1 \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \end{bmatrix}$$

where $\tan \theta := a$. (There is no specific physical meaning for the measurements of y_1 and y_2 but they are assumed here only for the convenience of discussion.) Then the transfer matrix for the spinning body can be computed as

$$Y(s) = P(s)U(s)$$

with

$$P(s) = \frac{1}{s^2 + a^2} \begin{bmatrix} s - a^2 & a(s + 1) \\ -a(s + 1) & s - a^2 \end{bmatrix}.$$

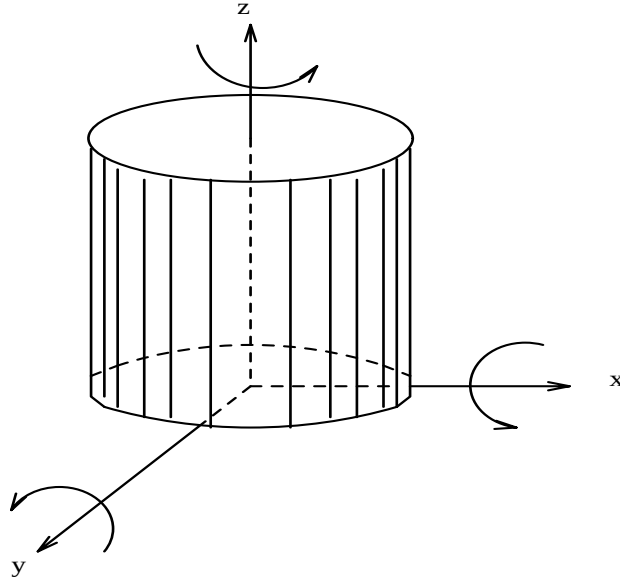


Figure 9.14: Spinning Body

Suppose the control law is chosen as

$$u = K_1 r - y$$

where

$$K_1 = \frac{1}{1+a^2} \begin{bmatrix} 1 & -a \\ a & 1 \end{bmatrix}.$$

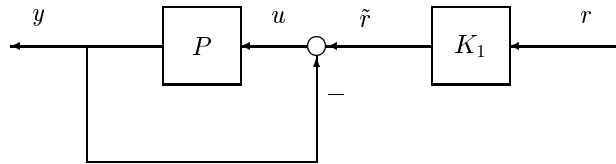


Figure 9.15: Closed-loop with a “Bizarre” Controller

Then the closed-loop transfer function is given by

$$Y(s) = \frac{1}{s+1} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} R(s)$$

and the sensitivity function and the complementary sensitivity function are given by

$$S = (I + P)^{-1} = \frac{1}{s+1} \begin{bmatrix} s & -a \\ a & s \end{bmatrix}, \quad T = P(I + P)^{-1} = \frac{1}{s+1} \begin{bmatrix} 1 & a \\ -a & 1 \end{bmatrix}.$$

It is noted that this controller design has the property of decoupling the loops. Furthermore, each single loop has the open-loop transfer function as

$$\frac{1}{s}$$

so each loop has phase margin $\phi_{\max} = -\phi_{\min} = 90^\circ$ and gain margin $k_{\min} = 0, k_{\max} = \infty$.

Suppose one loop transfer function is perturbed, as shown in Figure 9.16.

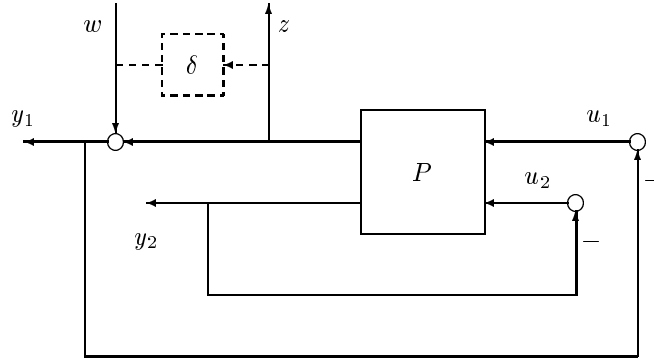


Figure 9.16: One-Loop-At-A-Time Analysis

Denote

$$\frac{z(s)}{w(s)} = -T_{11} = -\frac{1}{s+1}.$$

Then the maximum allowable perturbation is given by

$$\|\delta\|_\infty < \frac{1}{\|T_{11}\|_\infty} = 1$$

which is independent of a . Similarly the maximum allowable perturbation on the other loop is also 1 by symmetry. However, if both loops are perturbed at the same time, then the maximum allowable perturbation is much smaller, as shown below.

Consider a multivariable perturbation, as shown in Figure 9.17, i.e., $P_\Delta = (I + \Delta)P$, with

$$\Delta = \begin{bmatrix} \delta_{11} & \delta_{12} \\ \delta_{21} & \delta_{22} \end{bmatrix} \in \mathcal{RH}_\infty$$

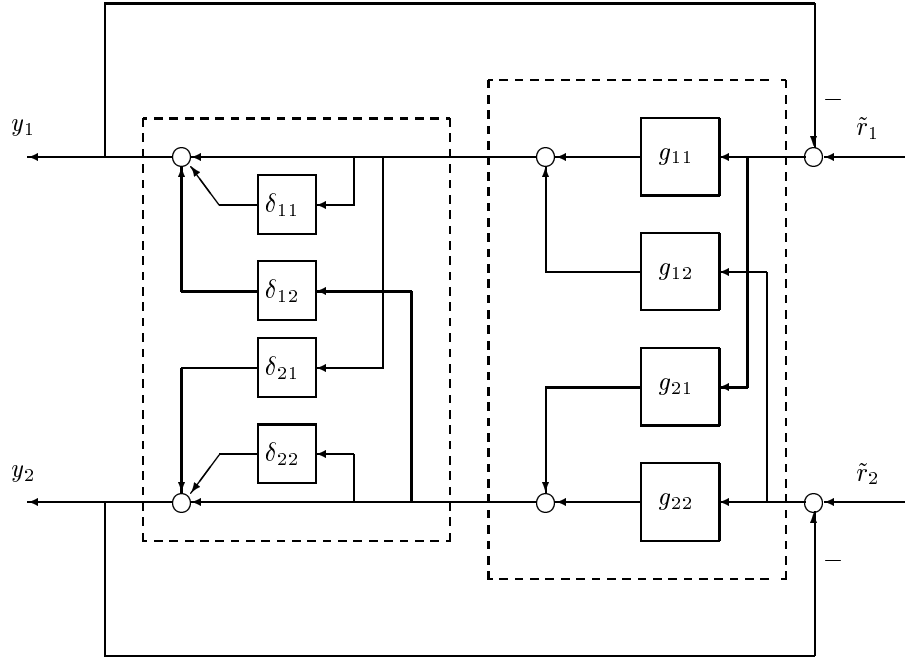


Figure 9.17: Simultaneous Perturbations

a 2×2 transfer matrix such that $\|\Delta\|_\infty < \gamma$. Then by the small gain theorem, the system is robustly stable for every such Δ iff

$$\gamma \leq \frac{1}{\|T\|_\infty} = \frac{1}{\sqrt{1+a^2}} \quad (\ll 1 \text{ if } a \gg 1).$$

In particular, consider

$$\Delta = \Delta_d = \begin{bmatrix} \delta_{11} & \\ & \delta_{22} \end{bmatrix} \in \mathbb{R}^{2 \times 2}.$$

Then the closed-loop system is stable for every such Δ iff

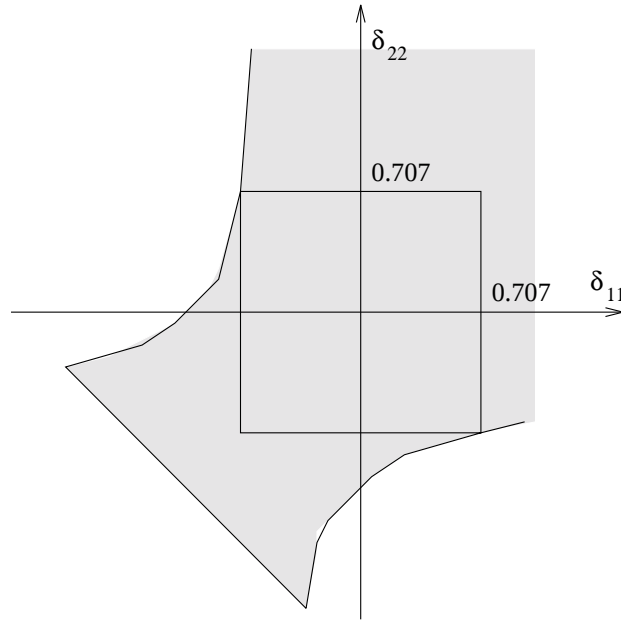
$$\det(I + T\Delta_d) = \frac{1}{(s+1)^2} (s^2 + (2 + \delta_{11} + \delta_{22})s + 1 + \delta_{11} + \delta_{22} + (1 + a^2)\delta_{11}\delta_{22})$$

has no zero in the closed right-half plane. Hence the stability region is given by

$$\begin{aligned} 2 + \delta_{11} + \delta_{22} &> 0 \\ 1 + \delta_{11} + \delta_{22} + (1 + a^2)\delta_{11}\delta_{22} &> 0. \end{aligned}$$

It is easy to see that the system is unstable with

$$\delta_{11} = -\delta_{22} = \frac{1}{\sqrt{1+a^2}}.$$

Figure 9.18: Stability Region for $a = 1$

The stability region is drawn in the Figure 9.18. This clearly shows that the analysis of an MIMO system using SISO methods can be misleading and can even give erroneous results. Hence an MIMO method has to be used.

9.7 Notes and References

The small gain theorem was first presented by Zames [1966]. The book by Desoer and Vidyasagar [1975] contains a quite extensive treatment and applications of this theorem in various forms. Robust stability conditions under various uncertainty assumptions are discussed in Doyle, Wall, and Stein [1982]. It was first shown in Kishore and Pearson [1992] that the small gain condition may not be necessary for robust stability with closed-ball perturbed uncertainties. In the same reference, some extensions of stability and performance criteria under various structured/unstructured uncertainties are given. Some further extensions are also presented in Tits and Fan [1994].

